

Vidhi Jain

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Enabling new capabilities with post-training in AI, robotics and chip design. Research scientist focused on imitation learning and RL post-training, with experience at OpenAI, Google DeepMind, Meta FAIR and CMU. Built training and evaluation pipelines, VLAs and video-conditioned robot policies published at RSS, CoRL, ICRA, and NeurIPS.

Education

Ph.D. in Robotics from Carnegie Mellon University (Robotics Institute).

Robotics Institute, Advisor: [Yonatan Bisk](#)

Masters of Science in Robotics, Carnegie Mellon University.

Robotics Institute, GPA 4.0 Advisor: [Katia Sycara](#)

Bachelor of Engineering (Honors) in Computer Science, BITS Pilani.

CGPA 9.06/10.00, Distinction, Advisor (off-campus thesis): [Aaron Courville](#)

Experience

Jul'25– **OpenAI, San Francisco, CA, USA.**

Mar'26 Researcher in AI for Science team with [Boris Power](#)

- Drove the capability improvement for frontier model for specification to code generation for chip design, utilizing reasoning models and verifiable rewards.
- Developed synthetic data pipelines to convert code into domain specific language for better verification and preventing reward hacking behaviors in RL post-training.

Jan'25– **Stanford.**

Jul'25 Collaborating with [Dorsa Sadigh](#) and [Jeanette Bohg](#)

- Designed ActVQ benchmark for an embodied visual querying environment where agents must locate fine-grained information (e.g., “find the expiry date” or “check ingredients”) under partial observability.

Jun'23– **Google DeepMind, Mountain View, CA, USA.**

Feb'24 Student Researcher with [Debidatta Dwibedi](#)

- Trained video-conditioned robot policies with cross-attention transformers. [Vid2Robot \(RSS'24\)](#)
- Extended models for visual grounding with conditional caption generation for QA. [FlexCap \(NeurIPS'24\)](#)
- Evaluated LLM prompts for low-level motion control for articulated object interaction. [PromptBook \(ICRA'24\)](#)
- Developed pipelines for zero-shot and fine-tuned task evaluations for single neural network policy trained on aggregated robotics data. [Open X-Embodiment \(ICRA'24\)](#) [Best Paper](#).

Sep'21– **Meta (Facebook), Menlo Park, CA, USA.**

Aug'22 AI resident with [Akshara Rai](#) and [Yixin Lin](#)

- Developed synthetic data for robot demonstrations of loading the simulated dishwasher in AI Habitat. [Code](#)
- Trained Encoder-decoder Transformer-based high-level policy for learning preferences in loading dishwasher from a single demonstration as prompt. [Paper](#) | [Website](#)

Aug'18– **Microsoft Research, Bengaluru, India.**

Aug'19 Research Fellow with [Amit Deshpande](#) and [Navin Goyal](#)

- **Divergence minimization in GANs** Investigated the properties of Jensen Shannon and Wasserstein divergences between the given and the generated distributions during training neural network generators, to discuss the learning dynamics like convergence and generalization.

Jan'18– **Montreal Institute for Learning Algorithms (MILA), Montreal, QC, Canada.**

Jun'18 Research Intern with [Aaron Courville](#)

- **Out-of-Distribution Detection in Generative Models** Experimented on autoregressive generative models and Variational Autoencoders, to detect any ‘out-of-distribution (OOD)’ sample at deployment. [Thesis](#)

- May'17– **Simon Fraser University (SFU)**, Burnaby, BC, Canada.
- Jul'17 MITACS Globalink Research Intern with [Oliver Schulte](#)
- **Learning Bayesian Networks for Relational Databases** Remodelled existing codebase for project *FactorBase* to integrate cross-table sufficient statistics or population variables in the contingency tables for learning ground-level sufficient statistics. [Code](#) | [Invited Paper](#)
- May'16– **Forschungszentrum Informatik (FZI)**, Karlsruhe, Germany.
- Jul'16 Research Intern with [York Sure-Vetter](#) and [Matthias Frank](#)
- **API annotation platform using Semantic Web** Designed and developed a customized Semantic MediaWiki for the decision support system of Project *BigGIS*; [Extended Abstract](#) | [Poster](#)

Selected Publications

- Dec 2024 **FlexCap: Generating Rich, Localized, and Flexible Captions in Images.**
Debidatta Dwibedi, **Vidhi Jain**, Jonathan Tompson, Andrew Zisserman, Yusuf Aytar. *Accepted at* Conference on Neural Information Processing Systems (NeurIPS) 2024, Vancouver, Canada. [Paper](#) | [Website](#)
- Aug 2024 **ANAVI: Audio Noise Awareness using Visuals of Indoors for NAVigation.**
Vidhi Jain, Rishi Veerapanini, Yonatan Bisk. *In Proceedings of* Conference of Robot Learning (CoRL) 2024, Munich, Germany. [Paper](#) | [Website](#)
- July 2024 **Vid2Robot: End-to-end Video-conditioned Policy Learning with Cross-Attention Transformers.**
Vidhi Jain, Maria Attarian, Nikhil J Joshi, Ayzaan Wahid, Danny Driess, Quan Vuong, Pannag R Sanketi, Pierre Sermanet, Stefan Welker, Christine Chan, Igor Gilitschenski, Yonatan Bisk, Debidatta Dwibedi, *In Proceedings of* Robotics: Science and Systems (RSS) 2024, Delft, Netherlands. [Paper](#) | [Website](#) | [Video](#)
- Apr 2024 **How to Prompt Your Robot: A PromptBook for Manipulation Skills with Code as Policies.**
Montserrat Gonzalez Arenas, Ted Xiao, Sumeet Singh, **Vidhi Jain**, Allen Z. Ren, Quan Vuong, Jake Varley, Alexander Herzog, Isabel Leal, Sean Kirmani, Dorsa Sadigh, Vikas Sindhwani, Kanishka Rao, Jacky Liang, Andy Zeng. *In Proceedings at* International Conference on Robotics and Automation (ICRA) 2024, Yokohama, Japan. [Paper](#)
- Nov 2023 **Open X-Embodiment: Robotic Learning Datasets and RT-X Models.**
Open X-embodiment collaboration. *In Proceedings at* International Conference on Robotics and Automation (ICRA) 2024, Yokohama, Japan. [Best Paper Paper](#) | [Website](#)
- Jun 2023 **SLAP: Spatial-Language Attention Policies.**
Priyam Parasher, **Vidhi Jain**, Xiaohan Zhang, Jay Vakil, Sam Powers, Yonatan Bisk, Chris Paxton. *In Proceedings of* Conference on Robot Learning (CoRL) 2023, Atlanta, USA., [Paper](#) | [Website](#)
- Jun 2023 **HomeRobot: Open-Vocabulary Mobile Manipulation.**
Sriram Yenamandra, Arun Ramachandran, Karmesh Yadav, Austin S Wang, Mukul Khanna, Theophile Gervet, Tsung-Yen Yang, **Vidhi Jain**, Alexander Clegg, John M Turner, Zsolt Kira, Manolis Savva, Angel X Chang, Devendra Singh Chaplot, Dhruv Batra, Roozbeh Mottaghi, Yonatan Bisk, Chris Paxton. *In Proceedings of* Conference on Robot Learning (CoRL) 2023, Atlanta, USA. [Paper](#) | [Website](#) | [Competition @ NeurIPS 2023](#)
- Dec 2022 **Transformers are Adaptable Task Planners.**
Vidhi Jain, Yixin Lin, Eric Undersander, Yonatan Bisk, Akshara Rai. *In Proceedings of* Conference on Robot Learning (CoRL) 2022, Auckland, New Zealand. [Paper](#) | [Website](#) | [Video](#) | [Code](#)

Workshop Presentations

- Dec 2022 **MAEA: Multimodal Attribution for Embodied AI.**
Vidhi Jain, Jayant Sravan Tamarapalli, Sahiti Yerramilli, Yonatan Bisk. *Presented at* NeurIPS 2022, 5th Robot Learning Workshop: Trustworthy Robotics (RLW 2022), New Orleans, USA. [Short Paper](#) | [Website](#) | [Video](#)
- Oct 2020 **Predicting Strategies in Simulated Search and Rescue.**
Vidhi Jain, Rohit Jena, Huao Li, Tejus Gupta, Dana Hughes, Michael Lewis, Katia Sycara. *Presented at* NeurIPS AI+HADR 2020, Virtual. [Video](#) | [Preprint](#)

- Oct 2020 **Learning Embeddings that Capture Spatial Semantics for Indoor Navigation.**
Vidhi Jain, Shishir Patil, Prakhar Agarwal, Katia Sycara. *Presented at NeurIPS 2020 Object Representations for Learning and Reasoning (ORLR 2020)*, Virtual. [Video](#) | [Preprint](#) | [Code](#)
- Sep 2020 **Learning to Navigate in Unseen Cluttered Structured Environments.**
Vidhi Jain, Ganesh Iyer, Katia Sycara. *NeurIPS 2020 Women In Machine Learning (WiML 2020)*, Virtual. [Poster](#)
- Jul 2020 **Coping with Sample Inefficiency in Deep Reinforcement Learning (DRL) in embodied AI.**
Vidhi Jain, Simin Liu, Ganesh Iyer. *ICML'20 Women In Machine Learning (WiML) Un-Workshop*, Virtual. [Discussion](#) | [Presentation](#)

Preprints

- Dec 2023 **Towards General-Purpose Robots via Foundation Models: A Survey and Meta-Analysis.**
Yafei Hu*, Quanting Xie*, **Vidhi Jain***, Jonathan Francis, Jay Patrikar, Nikhil Keetha, Seungchan Kim, Yaqi Xie, Tianyi Zhang, Shibo Zhao, Yu Quan Chong, Chen Wang, Katia Sycara, Matthew Johnson-Roberson, Dhruv Batra, Xiaolong Wang, Sebastian Scherer, Zsolt Kira, Fei Xia, Yonatan Bisk. [Paper](#) | [Website](#)
- Nov 2019 **Towards Zero-Shot Alignment and Retrieval for Forensic Detection.**
Vidhi Jain*, Amrit Setlur*, Aditya Pratapa*. [Report](#)

Tech Reports

- Aug 2025 **Lowering Barriers in Human-Robot Communication.** PhD Thesis. [Report](#)
- Aug 2021 **Towards Explainable Embodied AI.** Masters Thesis. [Report](#)
- Jun 2018 **Investigating the viability of Generative Models for Novelty Detection.** Bachelors Thesis. [Report](#)

Selected Awards and Honors

- Dec 2025 **BITS Global 30 Under 30 Award** in Research leaders – Academia category [Link](#)
- Aug 2020 **CIFAR Deep Learning Reinforcement Learning Summer School**
- Apr 2020 **Heidelberg Laureate Forum (HLF '20)** selected as a Young Researcher among 224 others worldwide.

Talks

- Mar 2024 **How to think of multimodality in Embodied AI and Robotic Manipulation?.**
Guest Lecture in Multimodal Machine Learning course. [Lecture Video](#)
- Apr 2023 **Training vision models in PyTorch.**
Hands-on lecture in Computer Vision course. [Lecture Video](#)
- Jan 2019 **The One in Asankhya Project.**
by Sukriti Paul and Mansi Goyal (Virtual) [Video](#) | [Blog](#)
- Jul 2017 **Invited panelist for 'Innovation by Young India'.**
telecasted on national news channel NDTV India at New Delhi office. [Video](#)

Professional Service and Leadership

- Sep'23– Present **Reviewer** at IEEE International Conference on Robotics and Automation (ICRA), International Conference on Intelligent Robots and Systems (IROS)
- Jan'20– Present **Reviewer** at International Conference on Machine Learning (ICML), Neural Information Processing and Signal (NeurIPS) and International Conference on Learning Representations (ICLR).
- Jun'20– Jul'20 **Workshop Session Organizer** | Coping with Sample Inefficiency in Deep Reinforcement Learning (DRL) in embodied AI, ICML'20 WiML Un-Workshop, Virtual. [Report](#) | [Presentation](#)